

Statistical Programming Languages

Exercises: Day 4

Exercise 1 (Basic Statistics):

The General Social Survey 2018 contains amongst others the following variables:

age	Age of Respondent
sex	Respondent's Sex (1=Male, 2=Female)
educ	Highest Year of School Completed
sibs	Number of brothers and sisters
life	Is life dull (=1), routine (=2) or exciting (=3)
speduc	Highest Year of School Completed by Spouse
paeduc	Highest Year of School Completed by Father
maeduc	Highest Year of School Completed by Mother
tvhours	Hours of Television Watched
wrkstat	Labor Force Status (1=Working fulltime)
hrs1	Number of Hours Worked Last Week
happy	General happiness (1=Very Happy, 2=Pretty happy, 3=Not too happy, NA=NAP/DK/NA)
hapmar	Happiness of marriage (coding as in happy)
degree	Respondent's highest degree
rincome	Respondent's income (in categories)
richwork	Respondent's attitude towards working if he/she was financially secure (1=Continue working, 2=Stop working)

Solve the following exercises. Use a significance level of 5% for all questions involving tests.

- Read in the General Social Survey 2018 from https://gss.norc.umd.edu/Documents/stata/2018_stata.zip. (Hint: Use the function `import()` from the package `rio`, compare slide 21 in 7.basic_statistics.pdf)
- Create appropriate plots describing the distributions of the variables `sibs` and `educ`. Which location parameter describes the distribution best?
- Consider respondent's income (`rincome`).
 - Examine the variable using the function `str()`. What do you observe? Transform the variable to class `factor` using reasonable labels.
 - Create a numeric variable `rincomdol` (respondent's income in US\$) where you replace the category value by the class center, e.g. \$1,000 to \$2,999 by the numeric value 2000. For the last category use 25000.
 - Compute the average respondent's income in US\$ for the different education levels (degree). What is the average income for all respondents? For people with less than a high school degree? For people with a graduate degree?

- (iv) Do you think that staying in school pays off?
- d) Consider the variable `age`.
 - (i) Create a histogram (with total area 1).
 - (ii) Overlay the histogram with the density of an appropriate normal distribution. Does the variable look normally distributed?
- e) Analyse the variable `age` with boxplots grouped by `life`. Transform the variable `life` to class factor with reasonable labels to do so.
 - (i) Which group has the largest dispersion? Which criterion do you use for dispersion?
 - (ii) Is the distribution in one or more groups symmetric?
 - (iii) What is the median in each group?
 - (iv) From the graphics: Do you believe there is a relationship between `age` and `life`?
- f) Consider the relationship between how happy a person is in general (`happy`) and the happiness of his or her marriage (`hapmar`).
 - (i) Transform both variables to class factor with reasonable labels and create a Mosaic plot for them. What do you learn?
 - (ii) Perform a Chi square independence test. Do the variables appear to be independent? Is this test appropriate for this table?
 - (iii) Compute Cramér's Φ , Cramér's V and the contingency coefficient(s). What can you say about the strength of the association between the two variables based on these values?
- g) Compute the correlation between `paeduc` (Highest Year of School Completed, Father) and `maeduc` (Highest Year of School Completed, Mother). What do we learn?
- h) Select full-time workers only (`wrkstat=1`).
 - (i) Create a scatterplot with `hrs1` and `educ`. Color the data points in the scatterplot according to `sex`.
 - (ii) Run simple linear regressions for both sexes together and separately for each sex with `hrs1` as dependent variable and `educ` as independent variable.
 - (iii) Add the regression lines to the scatter plot.
- i) Test whether Clinton and Trump supporters (`pres16`) differ in average age and education (`educ`).
- j) Some people claim that they would continue to work if they struck it rich, and others say that they would not (variable `richwork`). Use an appropriate test to identify possible differences of averages between these two groups in age (`age`), education (`educ`) and tv hours (`tvhours`).
- k) Run a regression model where a respondent's education depends on their spouse's education, father's education, respondent's income in dollars, and hours of television watched. Check the normality assumption.
- l) Write a short R Markdown report with your results from the exercises.